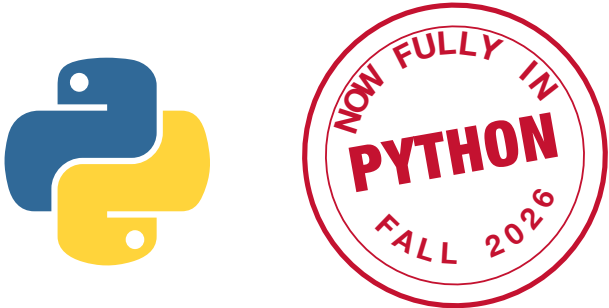


15-322 / 15-622 | FALL 2026

INTRODUCTION TO COMPUTER MUSIC

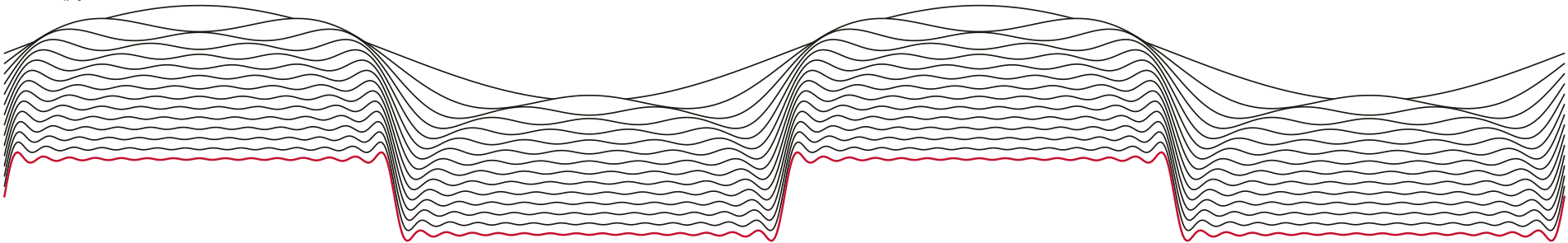
Turn code into sound.

Sampling, synthesis, Fourier analysis, effects, and generative music,
from first principles to real-time systems.



$$x_N(t) = \frac{4}{\pi} \sum_{n=1}^N \frac{\sin(2\pi(2n-1)f_0t)}{2n-1}$$

SQUARE WAVE PARTIAL SUMS · N = 1 TO 15



TUE + THU
11:00 AM – 12:20 PM



CIC 1203

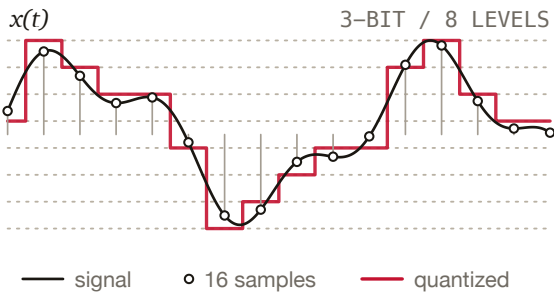


9 UNITS · 15-322
12 UNITS · 15-622

WHAT YOU’LL LEARN

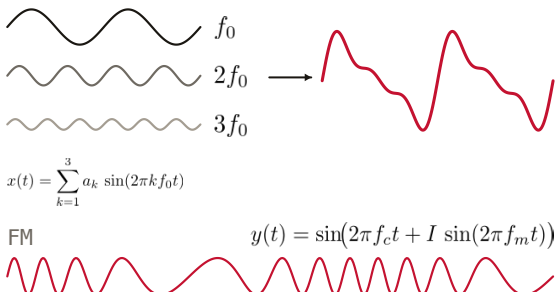
01 DIGITAL AUDIO

SAMPLING · QUANTIZATION · ALIASING



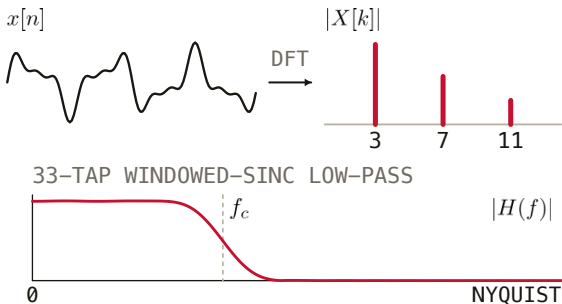
02 SYNTHESIS

ADDITIVE · FM · PHYSICAL MODELING



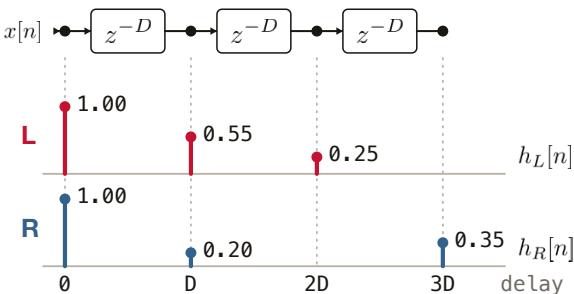
03 FOURIER & DSP

DFT / FFT · FILTERS · CONVOLUTION · STFT



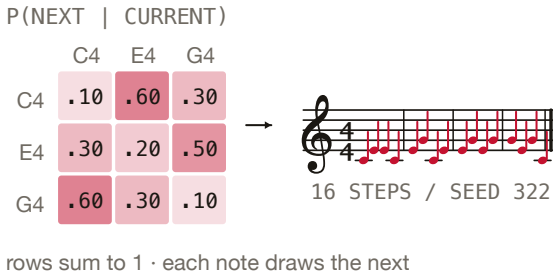
04 EFFECTS & SPATIALIZATION

DELAY LINES · MODULATION · PHASE VOCODING



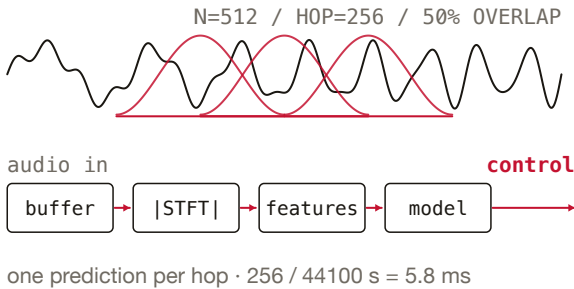
05 ALGORITHMIC COMPOSITION

MARKOV CHAINS · LPC · MIDI



06 REAL-TIME & MUSIC AI

STREAMING · MIR · GENERATIVE MODELS



OPEN-ENDED PROJECTS
EXPLORE YOUR OWN INTERESTS



PREREQUISITE
15-122 OR 15-112



MUSIC EXPERIENCE
NOT REQUIRED